C PROGRAMMING FROM SCRATCH · EPISODE 07

Comments

Explain the Code

KEY POINTS

- Comments explain source code for people; they add no program output.
- Use // to the end of a line, or /* and */ for a block.
- Block comments do not nest; prefix selected lines with // instead.
- Keep comments useful and current when the code changes.

01 Notes for people

▶ VIDEO 00:33

Comments are replaced with a space during preprocessing. They do not become instructions in the running program.



02 Two comment forms

▶ VIDEO 01:40

FORM	BOUNDARY	EXAMPLE
Line comment	`//` to the end of that line	`// Print the values`
Block comment	`/*` to the first `*/`	`/* Print the values */`

- A line comment can stand alone or follow a statement. It became standard in **C99**.
- A block comment can span one line or several lines. A newline does not close it.

03 Temporarily disable code

▶ VIDEO 02:41

Prefix each unwanted statement with `//`. Remove the markers to restore the code. Finish the experiment by restoring the code or removing what is no longer needed.

Excerpt: disabled-lines.c — C SOURCE

```c
    // printf("%d\n", 100);
    // printf("%c\n", 'A');
    printf("%f\n", 3.14159);
```

04 Why block comments cannot nest

▶ VIDEO 03:37

The **first closing marker** ends the comment, even if another opening marker appears inside it.

Invalid excerpt from nested-comments.c — C SOURCE

```c
/*
printf("%d\n", 100); /* integer */
printf("%c\n", 'A');
*/
```

The comment ends after `integer`. The second `printf` is still code, and the final `*/` is unmatched: this example fails to compile.

> WATCH OUT **Use one line-comment prefix per line**
>
> To disable statements that already contain a block comment, prefix each selected line with `//`. The existing `/* integer */` becomes line-comment text.

05 Visual Studio shortcuts

▶ VIDEO 04:44

Select **complete lines**, then press the two combinations in sequence.

COMMAND	FIRST	THEN
Comment Selection	**Ctrl+K**	**Ctrl+C**
Uncomment Selection	**Ctrl+K**	**Ctrl+U**

These are the default bindings. Customized keyboard layouts may differ.

06 Write comments worth keeping

▶ VIDEO 05:29

- Explain **why** a constraint or workaround exists, rather than repeating a statement.
- Learning notes can explain unfamiliar syntax. Improve unclear names before adding explanations of those names.
- When code changes, update its comments too.

Explains the constraint — C SOURCE

```c
// The test expects exactly 100.
printf("%d\n", 100);
```

CODE 7 files on GitHub github.com/abitclearer/c-programming-from-scratch/tree/main/07-comments